



DEEP LEARNING (MCTA 4363)

SECTION 1, SEMESTER 2, 2025/2026

PROJECT REPORT

TRACK 1 – BirdCLEF+ 2026

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1. Improvement from baseline

Iterative Model Development Log

Version	Model Strategy	Key Specs	CV / LB Score	Notes
v1.0	Perch v2 (Zero-Shot)	5s windows, 32kHz, Taxonomy Mapping	LB: 0.825	High-performance baseline using transfer learning.
v1.1	Fine-tuning (Planned)	Custom Linear Head for 234 species	TBD	Focused on missing Pantanal labels.

2. Implementation & Techniques

A. Foundation Model Selection:

We selected **Google Perch v2** (EfficientNet-B3 backbone) as our foundation. This model is pre-trained on a massive global dataset, making it robust for extracting features from various bird vocalizations.

B. Taxonomy Alignment

A critical component of our pipeline is the **Taxonomy Mapping**. We merged the competition's taxonomy.csv with the model's internal labels.csv to ensure that model outputs correctly correspond to the 234 target species required for the submission.

C. Efficient Processing:

To ensure the model executes within the Kaggle environment's time and resource constraints, we implemented several efficiency protocols:

- **Multi-threaded Inference:**

Utilizing ThreadPoolExecutor allowed us to overlap audio I/O tasks with model prediction, significantly reducing overall latency.

- **Temporal Watchdog Timer:**

We implemented a TERMINATE_TIME logic that triggers a graceful exit and CSV generation before the 2-hour environment timeout, preventing submission failures.

- **TensorFlow Optimizations:**

We utilized tf.experimental.numpy and tf.pad to efficiently handle missing species indices and ensure the submission format remains valid.

3. Insights Gained (Critical Analysis)

- **The Coverage Gap:**

Our analysis identified that the pre-trained Perch model only covers a portion of the 234 species required for this competition. This "zero-shot" limitation creates a performance ceiling, as the model cannot predict species it was never trained on, impacting our macro-averaged ROC-AUC score.

- **Temporal Precision:**

We evaluate the model using 5-second windows(5 x sr). While this is a standard industry window, bird calls in the Pantanal are often short bursts. This window size may overlap multiple species or capture irrelevant background noise, suggesting a need for finer temporal resolution in future iterations.

- **Environmental Noise robustness:**

The Pantanal soundscapes are dense with non-target sounds like amphibians and insects. We observed that the Perch model's multi-taxa training helps it filter these sounds, focusing primarily on the bird vocalizations of interest.

4. Leaderboard Standing & Repository

- **Current Standing:** 0.825 (Top 2190 as of 5 May 2026).
- **GitHub Repository:** <https://github.com/hafiz-fansuri/Deep-Learning>